

Mode-Path Matrix Quick Reference

Thermal Photonics & Diffusionics — Chapter 4 Companion

Thermal Photonics: A Unified Framework for Engineering Heat Flux

Book 3 of the Open-Source Engineering Optics Trilogy

Version 4.1 | Standalone practitioner reference

What To Do First

Standalone desk reference for the Mode-Path Matrix methodology. For full derivation see Chapter 4. Work through the embedded WE-4.0 condensed example (§1.7) before tackling real systems.

New in v4.0/4.1: Self-consistent iteration (§4); edge schema (§3.5); sensitivity coefficients (§2.1); dual MPM for quantum systems (§11); near-field G_{rad} enhancement (§1.3); linearization decision rule (§1.4); expanded failure modes (§12); semiconductor mini-example (§9); optics minimum-viable MPM (§10); units sanity table (§1.2); matrix fill rules (§3.4); DR numbering legacy map (§16).

1 The Thermal Network Paradigm

Any thermal system can be represented as a graph $\mathcal{G}(V, E)$:

- **Vertices V :** thermal nodes (sources, sinks, junctions).
- **Edges E :** thermal conductances G [W/K] connecting nodes.

Conduction edges are *undirected* ($G_{ij} = G_{ji}$). Convection edges carry a flow direction (directed). Radiation edges are symmetric by view-factor reciprocity ($\varepsilon_i A_i F_{ij} = \varepsilon_j A_j F_{ji}$).

1.1 Conductance Formulae

$$G_{\text{cond}} = \frac{\kappa A}{L}, \quad G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}}\bar{T}^3 A, \quad G_{\text{conv}} = hA, \quad G_{\text{contact}} = \frac{A}{R_c} \quad (1)$$

1.2 Units and Scaling Sanity Check

Table 1: Typical parameter ranges for quick order-of-magnitude sanity checks.

Parameter	Typical range	Unit	Example
κ (bulk)	0.2–400	W/(m·K)	Si: 150; Cu: 400; SiO ₂ : 1.4
κ (TIM)	1–5	W/(m·K)	Thermal grease: 3
ε (metals, polished)	0.02–0.1	—	Au: 0.03; Al: 0.05
ε (oxides, polymers)	0.7–0.95	—	SiO ₂ : 0.9; black paint: 0.95
h (natural conv.)	5–25	W/(m ² ·K)	Vertical plate in air
h (forced conv.)	50–500	W/(m ² ·K)	Fan-cooled heat sink
A (die/chip)	10 ^{−5} –10 ^{−3}	m ²	10×10 mm ² = 10 ^{−4}
L (substrate)	10 ^{−4} –10 ^{−2}	m	0.8 mm wafer = 8×10 ^{−4}

Quick check: For a Si substrate ($\kappa = 150$, $A = 10^{-4}$ m², $L = 8 \times 10^{-4}$ m): $G_{\text{cond}} = 150 \times 10^{-4} / 8 \times 10^{-4} = 18.75$ W/K. At 300 K with $\varepsilon = 0.7$: $G_{\text{rad}} = 4 \times 0.7 \times 5.67 \times 10^{-8} \times 300^3 \times 10^{-4} = 4.3 \times 10^{-4}$ W/K. So $\Gamma = 2.3 \times 10^{-5}$: purely conduction-dominated at room temperature.

1.3 Near-Field Radiation Enhancement

Eq. (1) gives the *far-field* radiation conductance, valid for gap $d \gg \lambda_{\text{th}} \approx 10 \mu\text{m}$ at 300 K. When $d < \lambda_{\text{th}}$, evanescent coupling enhances transfer (Ch. 3):

$$G_{\text{rad}}^{\text{NF}} \approx G_{\text{rad}}^{\text{FF}} \times \left(\frac{\lambda_{\text{th}}}{d} \right)^2 \quad (d \ll \lambda_{\text{th}}) \quad (2)$$

In 3D IC stacks with $d \sim 1 \mu\text{m}$, G_{rad} can exceed the far-field estimate by 10^2 – 10^4 , shifting a path from conduction to mixed-mode.

Design Rule:

DR-T.1c — Near-Field Check For gap $d < 10 \mu\text{m}$ between parallel surfaces at $T > 200 \text{ K}$: compute $G_{\text{rad}}^{\text{NF}}$ via Eq. (2) or Ch. 3 Eq. 03.18. If Γ crosses 0.1, reclassify as mixed-mode.

1.4 Linearization Decision Rule (DR-T.1b)

Design Rule:

DR-T.1b — Linearization Guard **If** $|\Delta T|/\bar{T} < 0.2$: use linearized $G_{\text{rad}} = 4 \varepsilon \sigma_{\text{SB}} \bar{T}^3 A$.
Else: use exact $Q_{\text{rad}} = \varepsilon \sigma_{\text{SB}} A (T_1^4 - T_2^4)$ inside Algorithm 09.A (§4), defining an effective $G_{\text{rad}}^{\text{eff}} = Q_{\text{rad}}/\Delta T$.

Numeric counterexample: Two plates at $T_1 = 500 \text{ K}$, $T_2 = 300 \text{ K}$, $\varepsilon = 0.8$, $A = 0.01 \text{ m}^2$. Mean temperature $\bar{T} = 400 \text{ K}$; $\Delta T/\bar{T} = 200/400 = 0.5 > 0.2$ — linearization is *not* valid.

Linearized: $G_{\text{rad}}^{\text{lin}} = 4 \times 0.8 \times 5.67 \times 10^{-8} \times 400^3 \times 0.01 = 0.116 \text{ W/K}$; $Q^{\text{lin}} = 0.116 \times 200 = 23.2 \text{ W}$.

Exact: $Q^{\text{exact}} = 0.8 \times 5.67 \times 10^{-8} \times 0.01 \times (500^4 - 300^4) = 0.8 \times 5.67 \times 10^{-8} \times 0.01 \times 5.44 \times 10^{10} = 24.7 \text{ W}$.

Error: $(24.7 - 23.2)/24.7 = 6.1\%$. At $\Delta T/\bar{T} = 0.5$ the linearization under-predicts by $>6\%$. For $\Delta T/\bar{T} > 1$ (e.g., 300 K to 1000 K), the error exceeds 30%.

1.5 Thermal Kirchhoff's Laws

Valid under: (1) lumped-parameter ($\text{Bi} < 0.1$); (2) linear constitutive $Q = G \cdot \Delta T$ (see §1.4 for radiation); (3) heat generation at nodes.

TKCL: $\sum_j G_{ij}(T_j - T_i) + Q_i = 0$ (steady state); replace 0 with $C_i dT_i/dt$ for transient. **TKVL:** $\sum_{\text{loop}} \Delta T_k = 0$.

1.6 Matrix Formulation

$$\mathbf{G} \cdot \mathbf{T} = \mathbf{Q} \quad (\text{steady state}), \quad \mathbf{C} \frac{d\mathbf{T}}{dt} + \mathbf{G} \cdot \mathbf{T} = \mathbf{Q}(t) \quad (\text{transient}) \quad (3)$$

Conduction Mode (Diffusionics)**Matrix Fill Rules** (for N unknown-temperature nodes):

- **Diagonal:** $G_{ii} = \sum_{j \neq i} G_{ij}$ (sum of all conductances connected to node i).
- **Off-diagonal:** $G_{ij} = -G_{ji}$ (negative of the conductance between nodes i and j).
- **Fixed- T (Dirichlet) nodes:** Do *not* assign a row/column. Move $G_{ij}T_{j,\text{fixed}}$ to the right-hand side $\mathbf{Q}$.
- **Heat-flux (Neumann) nodes:** Add Q_i directly to $\mathbf{Q}_i$.

Result: $\mathbf{G}$ is symmetric positive-definite (for conduction); solve $\mathbf{T} = \mathbf{G}^{-1}\mathbf{Q}$.**1.7 Condensed WE-4.0: Four-Node Starter Example****Case Study:**

WE-4.0 — Die–TIM–Cu–Sink **Setup:** Die (50 W), TIM, Cu substrate, heat sink (fixed at $T_4 = 25^\circ\text{C}$). Conductances: $G_{12} = 2 \text{ W/K}$ (Die–TIM); $G_{23} = 50 \text{ W/K}$ (TIM–Cu); $G_{34} = 10 \text{ W/K}$ (Cu–Sink).

Assemble 3×3 matrix (node 4 is Dirichlet):

$$\begin{pmatrix} 2 & -2 & 0 \\ -2 & 52 & -50 \\ 0 & -50 & 60 \end{pmatrix} \begin{pmatrix} T_1 \\ T_2 \\ T_3 \end{pmatrix} = \begin{pmatrix} 50 \\ 0 \\ 10 \times 25 \end{pmatrix} = \begin{pmatrix} 50 \\ 0 \\ 250 \end{pmatrix}$$

Node 3 RHS includes $G_{34} \times T_4 = 250 \text{ W}$.**Solution:** $T_1 = 55.1^\circ\text{C}$, $T_2 = 30.1^\circ\text{C}$, $T_3 = 29.3^\circ\text{C}$.**Critical path:** Die→TIM→Cu→Sink. $R_{\text{total}} = 1/2 + 1/50 + 1/10 = 0.50 + 0.02 + 0.10 = 0.62 \text{ K/W}$.**Bottleneck:** TIM ($R = 0.50 \text{ K/W}$), $\beta = 0.50/0.62 = 0.81$. Fix the TIM first (DR-T.3).**Mode check:** All edges at $\sim 300 \text{ K}$ with solid interfaces: $\Gamma < 10^{-4}$ everywhere—purely conduction-dominated. Use Part II tools.**2 Critical Path Analysis****Critical path:** source-to-sink route with highest total thermal resistance:

$$R_{\text{critical}} = \max_p \left(\sum_{e \in p} G_e^{-1} \right) \quad (4)$$

Bottleneck ratio:

$$\beta = R_{\text{bottleneck}}/R_{\text{critical}} \quad (5)$$

Table 2: Bottleneck ratio β interpretation.

β	Interpretation	Action
> 0.5	Single dominant bottleneck	Fix this element
$0.2\text{--}0.5$	Moderate bottleneck	Fix, then reassess
< 0.2	Distributed resistance	System-level optimization

2.1 Sensitivity Coefficients for Manufacturing

Quantify how fabrication parameter p_i affects system performance Φ via the **sensitivity coefficient**¹:

$$S_i = \frac{\partial \ln \Phi}{\partial \ln p_i} = \frac{p_i}{\Phi} \frac{\partial \Phi}{\partial p_i} \quad (6)$$

Table 3: Sensitivity classification and tolerance (DR-13.1, DR-13.2).

$ S_i $	Class	Tolerance	Metrology
> 0.5	Critical	$\pm 1\text{--}2\%$	100% inspection
$0.1\text{--}0.5$	Secondary	$\pm 5\text{--}10\%$	Lot sampling
< 0.1	Non-critical	$\pm 20\%$	First-article only

β tells you *where* to improve; S_i tells you *what to control* during fabrication. See Chapter 13.

3 The Mode-Path Matrix

3.1 Mode Dominance Ratio

$$\Gamma = G_{\text{rad}}/G_{\text{cond}} \quad (7)$$

Table 4: Mode classification via Γ .

Γ	Mode	Toolkit	Book Ref.
< 0.1	Conduction	Diffusionics	Part II (5–7)
$0.1\text{--}10$	Mixed	Multiphysics	Ch. 9
> 10	Radiation	Thermal Photonics	Part III (8–10)
$\rightarrow 0$ cryo	Conduction	Cryogenic/quantum	Ch. 14

Single-mode error (Ch. 9, Eq. 09.51): $\Delta R/R = \Gamma/(1 + \Gamma)$. At $\Gamma = 0.1$: 9% error; at $\Gamma = 1$: 50%; at $\Gamma = 10$: 91%.

Γ **landscape**: SC qubits 20 mK: $< 10^{-12}$; room-temp IC: $\sim 10^{-2}$; PECVD 400 °C: $\sim 1\text{--}5$; RTP 1100 °C: > 50 . See Figure 14.11.

3.2 MPM Fill-Out Recipe

What To Do First

Row-by-row procedure:

1. **List every path** source→sink (DR-T.1). One row per path.
2. **Compute** $G_{\text{cond}} = \kappa A/L$ (sanity-check with Table 1).
3. **Compute** G_{rad} (linearized). Check DR-T.1b (§1.4). If gap $d < 10 \mu\text{m}$: DR-T.1c (§1.3).
4. **Compute** $G_{\text{conv}} = h A$ if present (§3.6).
5. **Compute** Γ ; classify (Table 4).
6. **Flag mixed-mode** ($0.1 < \Gamma < 10$): iteration required (§4).
7. **Map to tool** (Table 8).

¹Denoted β_i in Chapter 13. Renamed to S_i here to avoid confusion with the bottleneck ratio β .

3.3 Template Mode-Path Matrix

Table 5: Template MPM (v4.1).

Path	From	To	G_{cond} (W/K)	G_{rad} (W/K)	G_{conv} (W/K)	Γ	Iter.	Tool
P1	Source	Spreader	50	0.5	—	0.01	No	Ch. 5–7
P2	Spreader	Ambient	2	8	5	4.0	Yes	Ch. 9
P3	Source	Lid	10	3	—	0.3	Yes	Ch. 9
P4	Lid	Ambient	1	5	20	5.0	No	BC only

3.4 Matrix Fill Rules

When populating the Mode-Path Matrix (Table 5), observe the following rules:

1. **One row per distinct source→sink path.** Parallel paths between the same node pair get separate rows; their conductances are *not* pre-summed.
2. **Evaluate G_{cond} and G_{rad} at the operating temperature $\bar{T}$,** not at 300 K. For G_{rad} , verify DR-T.1b ($\Delta T/\bar{T} < 0.2$).
3. **Include convective edges** ($G_{\text{conv}} = hA$) as a separate column; do not lump into G_{cond} .
4. **Leave no cell blank.** If a mode is absent on an edge, enter 0; this makes $\Gamma = 0$ or $\Gamma \rightarrow \infty$, both of which are valid classifications.
5. **Record units consistently** (W/K for all conductances; dimensionless for Γ).

3.5 Advanced Edge Schema (DR-09.8)

For production documentation of mixed-mode paths, each edge carries (Ch. 9, §09.7.1):

Table 6: Edge Schema fields (DR-09.8).

Field	Description
Edge ID	Node i –Node j
G_{cond} (W/K)	Conduction conductance
$G_{\text{rad}}(T_{\text{ref}})$	Radiation cond. at T_{ref}
ε_{eff}	Effective emissivity
F_{ij}	View factor
A (m ²)	Radiating area
$\Gamma(T_{\text{ref}})$	Mode dominance ratio
Spectral band	Wavelength range for ε
ω_{rec}	Under-relaxation (Table 7)
δG_{rad} (W/K)	Uncertainty
Convergence	Alg. 09.A needed? (Yes/No)

Example (WE-09.3, Wafer–Showerhead): $G_{\text{cond}} = 0.189$; $G_{\text{rad}}(673\text{ K}) = 0.32$ W/K; $\varepsilon_{\text{eff}} = 0.26$; $F_{14} = 0.25$; $A = 0.0707$ m²; $\Gamma = 1.7$; LWIR 8–14 μm ; $\omega_{\text{rec}} = 0.5$; $\delta G_{\text{rad}} = \pm 0.05$ W/K; Yes, ~ 6 iterations.

3.6 Convection in the MPM

Convection is *not* a third microscopic mode but a boundary-layer conductance. The MPM records G_{conv} but routes design via Γ alone.

1. **Boundary-only:** G_{conv} in BC column. Design uses Γ .

2. **Internal forced flow:** $G_{\text{eff}} = \dot{m} c_p$. Path stays $\Gamma \ll 1$.

Pain Point

Convection-dominated path ($G_{\text{conv}} \gg G_{\text{cond}}$ and $G_{\text{conv}} \gg G_{\text{rad}}$): This path is **not a metamaterial design target**. Mark “**BC only**” in the Tool column. Optimize h or flow rate using conventional fluid-mechanics tools. See Appendix D.

4 Self-Consistent Iteration (Algorithm 09.A)

For mixed-mode paths ($0.1 < \Gamma < 10$), single-pass $G_{\text{total}} = G_{\text{cond}} + G_{\text{rad}}(T_0)$ fails because $G_{\text{rad}} \propto T^3$ creates feedback (Ch. 9, §09.5):

Multiphysics Integration

Algorithm 09.A — Coupled Solver:

1. **S1: Initialize.** $\mathbf{G}^{\text{cond}} \mathbf{T}^{(0)} = \mathbf{Q} + \mathbf{b}$.
2. **S2: Compute radiation.** $G_{\text{rad}}^{(k)} = 4 \varepsilon_{\text{eff}} F_{ij} \sigma_{\text{SB}} (T_{ij}^{(k)})^3 A$.
3. **S3: Solve coupled.** $[\mathbf{G}^{\text{cond}} + \mathbf{G}^{\text{rad},(k)}] \tilde{\mathbf{T}}^{(k)} = \mathbf{Q} + \mathbf{b}$.
4. **S4: Under-relax.** $\mathbf{T}^{(k+1)} = \omega \tilde{\mathbf{T}}^{(k)} + (1 - \omega) \mathbf{T}^{(k)}$.
5. **S5: Check.** $r_T < \epsilon_T$ and $r_E < \epsilon_E$.
6. **S6: Stall?** $k > N_{\text{max}}/2$ and $\rho > 0.9 \rightarrow$ Newton–Raphson. Flag at $N_{\text{max}} = 50$.
7. **S7: Post-process.** Converged Γ ; update MPM.

Defaults: $\epsilon_T = 0.01$ K; $\epsilon_E = 10^{-3} Q_{\text{max}}$; $N_{\text{max}} = 50$. Typical: 3–15 iterations.

Table 7: Under-relaxation ω vs. Γ (Ch. 9, Table 09.7). Note: $\omega_{\text{max}} > 1$ at low Γ represents over-relaxation; always use ω_{rec} unless you have verified convergence.

Γ	ω_{max}	ω_{rec}	Typical iter.
0.1	3.7	0.7	3–4
0.5	1.0	0.6	4–5
1.0	0.67	0.5	5–6
2.0	0.50	0.4	6–8
5.0	0.40	0.3	8–12
10	0.37	0.3	10–15

5 Tool Translation Table

Table 8: Rosetta Stone: Γ to design action.

Γ	Physics	Tools	Ch.	Recipe
< 0.1	Fourier; $\mathbf{G} \cdot \mathbf{T} = \mathbf{Q}$	COMSOL, Icepak, FloTHERM	5–7	Min. R_{crit} ; improve $\kappa A/L$.
> 10	Stefan–Boltzmann; radiosity	Code V, Zemax, LightTools	8–10	Modify ε ; spectral shield.
0.1–10	Coupled	COMSOL+rad.;9 MPM-guided		Alg. 09.A; iterate.
$\rightarrow 0$ cryo	Kapitza; $G_{\text{rad}} \propto T^3 \rightarrow 0$	Cryogenic FEA	14	Maximize parasitic R .

Spectral sub-routing ($\Gamma > 10$): Broadband $\varepsilon \rightarrow$ Ch. 8; narrowband selective emitter (TPV) $\rightarrow$ Ch. 10; radiative cooling/shielding $\rightarrow$ Ch. 9 (§09.1–§09.3).

6 Three Limiting Cases

1. **Photon-only** ($\Gamma \gg 1$ all paths): MPM $\rightarrow$ radiosity. Ray-trace. Ex: furnace.
2. **Phonon-only** ($\Gamma \ll 1$ all paths): MPM $\rightarrow R$ -network. FEA. Ex: IC at 25 °C.
3. **Mixed**: Full MPM needed. Ex: fab 25 °C–1100 °C.

The MPM’s first value is *telling you which case you are in*.

7 Conductance Matching

Mismatch ratio for two series elements:

$$M = G_{\text{high}}/G_{\text{low}} = R_{\text{low}}/R_{\text{high}} \quad (8)$$

Table 9: Mismatch severity.

M	Severity	Consequence
1–3	Well-matched	Efficient
3–10	Moderate	10–30% loss
10–100	Severe	Bottleneck-dominated
> 100	Extreme	One element irrelevant

8 System Design Workflow: Six Steps

Multiphysics Integration

Thermal Architecture Design Process:

1. **Define requirements.** $T_{j,\max}$, power map, $\Delta T_{\max}$, τ_{proc} .
2. **Construct network.** All nodes/paths; initial G (DR-T.1). Verify $\text{Bi} < 0.1$.
3. **Find critical path.** Topology (DR-T.2); highest- R route (DR-T.3); bottleneck β .
4. **Build MPM.** Γ for each path (DR-T.4). Check DR-T.1b, DR-T.1c. Use §3.2.
5. **Apply tools.** Cond. $\rightarrow$ Part II. Rad. $\rightarrow$ Part III. Mixed $\rightarrow$ Ch. 9 + Alg. 09.A. Check M (DR-T.5).
6. **Verify and iterate.** Recalculate—critical path may shift (DR-T.6). Repeat.

9 Semiconductor Worked Mini-Example

Case Study:

PECVD Chamber at 400 °C Nodes: Plasma ($Q = 500$ W), Wafer, Chuck, Coolant (20 °C).

Path	Nodes	G_{cond} (W/K)	G_{rad} (W/K)	Γ	Mode
P1	Plasma→Wafer	0.19	0.32	1.7	Mixed
P2	Wafer→Chuck	5.0	0.03	0.006	Cond
P3	Chuck→Coolant	50	0	0	Cond
P4	Plasma→Chamber	0.05	1.2	24	Rad

Analysis:

1. Critical path: P1–P2–P3. **Key point:** for mixed-mode P1, use $G_{\text{total}} = G_{\text{cond}} + G_{\text{rad}} = 0.19 + 0.32 = 0.51$ W/K (not G_{cond} alone—this is the error that failure mode #8 warns about). $R_{\text{total}} = 1/0.51 + 1/5.03 + 1/50 = 1.96 + 0.20 + 0.02 = 2.18$ K/W.
2. Bottleneck: P1, $\beta = 1.96/2.18 = 0.90$ (dominant).
3. P1 is mixed ($\Gamma = 1.7$) $\rightarrow$ Alg. 09.A with $\omega = 0.5$, ~ 6 iterations.
4. P4 is radiation-dominated ($\Gamma = 24$): parasitic path. Add shield (Ch. 9) if needed.
5. After convergence, update G_{rad} on P1 with self-consistent value. Recheck critical path (DR-T.6).

10 Optics Minimum-Viable MPM

Trilogy Connection:

Default Node/Edge List for Optical Hardware For any optical instrument (imaging system, spectrometer, laser cavity), this five-node template captures the dominant thermal paths. Fill in your system-specific G values and compute Γ ; the result tells you immediately whether thermal-optical coupling matters.

	From	To	Typical Γ	Often missed?
E1	Lens	Mount (conduction)	< 0.01	No
E2	Lens	Housing (radiation)	0.01–1	Yes
E3	Housing	Ambient (conv.+rad.)	0.5–5	No
E4	Detector	Mount (conduction)	< 0.01	No
E5	Detector	Housing (radiation)	0.01–0.5	Sometimes
E6	Source/laser	Mount	< 0.1	No

E2 is the classic “missing path” (Failure Mode #1): the radiative exchange between a warm lens and the housing is often neglected in conduction-only thermal models. At $T \sim 50^\circ\text{C}$ with $\varepsilon \sim 0.9$ (coated glass), Γ for E2 can reach ~ 0.3 —mixed mode. This causes asymmetric thermal gradients across the lens, producing wavefront error (Eq. 9).

Optical consequence: For each edge through a refractive element, add a WFE column: $\text{WFE} \approx (dn/dT) \cdot \Delta T \cdot L/\lambda$. If any $\text{WFE} > \lambda/10$, the thermal design must be tightened (Book 1 connection).

11 Dual MPM for Quantum Systems (Ch. 14)

Quantum Extension:

DR-14Q.1 — Dual Network Optimization Construct **two** Mode-Path Matrices:

- **Cooling MPM:** intentional paths (minimize R_{critical}).
- **Parasitic MPM:** heat leaks (maximize $R_{\text{parasitic}}$).

Optimize both simultaneously. At $T < 1\text{ K}$, G_{rad} is suppressed by > 12 orders of magnitude vs. 300 K ; all internal paths $\Gamma < 10^{-6}$. Exception: stray IR through viewports ($\Gamma \gg 10$).

12 Common MPM Failure Modes

Failure Mode:

Top Eight Practitioner Mistakes

1. **Missing paths (DR-T.1).** Forgetting radiative paths (lens-to-housing, §10 edge E2) or parasitic conduction (cables). Prevention: systematic enumeration.
2. **Mode misidentification.** Assuming conduction-only when $\Gamma = 2.5$. CS-6 in Ch. 12: 40% error. Prevention: compute Γ numerically.
3. **Linearization error (DR-T.1b).** Using linearized G_{rad} when $\Delta T/\bar{T} > 0.2$. See §1.4 counterexample. Prevention: check ratio for every edge.
4. **Optimizing wrong element (DR-T.5).** Via array $G = 1000$ when TIM $G = 1$ is bottleneck ($M = 1000$). Prevention: compute M first.
5. **Not re-checking (DR-T.6).** Critical path shifts after fix (CS-5, Ch. 12). Prevention: rebuild MPM after every change.
6. **Near-field underestimate (DR-T.1c).** Far-field G_{rad} for gaps $d < 10 \mu\text{m}$. True G_{rad} may be 10^2 – $10^4 \times$ higher. Prevention: check d vs. λ_{th} .
7. **Observer effect (DR-13.10).** Probes perturb the network. Prevention: $\eta_{\text{obs}} = G_{\text{meas}}/G_{\text{crit}} < 5\%$. Non-contact mandatory for $T < 10 \text{ K}$.
8. **Skipping iteration (DR-09.5).** Single-pass $G_{\text{total}} = G_{\text{cond}} + G_{\text{rad}}(T_0)$ instead of Alg. 09.A. Off by 10–30%. Prevention: always iterate when $0.1 < \Gamma < 10$. See §9 for the correct treatment.

13 Navigation Routing Guide

- $\Gamma < 0.1 \rightarrow$ Part II. Ch. 5 (transform/EMT); Ch. 6 (switchable); Ch. 7 (via arrays).
- $0.1 < \Gamma < 10 \rightarrow$ Ch. 9 + Alg. 09.A.
- $\Gamma > 10 \rightarrow$ Part III. Ch. 8 (spectral); Ch. 9 (coupled); Ch. 10 (TPV).
- $\Gamma \rightarrow 0 \text{ cryo} \rightarrow$ Ch. 14. Dual MPM (§11).
- Convection-limited $\rightarrow$ Appendix D (§3.6).

14 Robustness, Temporal Compatibility, and Optical Bridge

Design Rule:

DR-T.7 — Manufacturing Robustness $G_{\text{design}} > G_{\text{required}} \times (1 + 3 \times \text{CV})$, $\text{CV} = 0.05$ – 0.15 .

Design Rule:

DR-T.8 — Temporal Compatibility $\tau_{\text{th}} < \tau_{\text{proc}}$. If violated, transient analysis (Eq. 3) is mandatory.

Trilogy Connection:Thermal $\rightarrow$ Optical Error

$$\text{WFE} \approx \frac{(dn/dT) \cdot \Delta T \cdot L}{\lambda} \quad [\text{waves}] \quad (9)$$

If $\text{WFE} > \lambda/10$, tighten thermal design. Every MPM path through a refractive element should carry a WFE estimate. Connects to Book 1 (*The Eikonal Bridge*) and Book 2 (*Quantum Field Imaging*).

15 Complete Design Rules Summary

Table 10: Topological design rules (DR-T series). DR-T.1c new in v4.0.

Rule	Key	Description	Eq.
DR-T.1	All paths	Complete network. $\text{Bi} < 0.1$.	—
DR-T.1b	$\Delta T/\bar{T}$	< 0.2 for linearized G_{rad} . See §1.4.	(1)
DR-T.1c	Near-field	Gap $< 10 \mu\text{m}$: compute $G_{\text{rad}}^{\text{NF}}$; reclassify if Γ crosses 0.1.	(2)
DR-T.2	Topology	Classify before component selection.	—
DR-T.3	Highest R	Critical path; β .	(4)
DR-T.4	Γ	MPM for <i>all</i> paths. Most important DR.	(7)
DR-T.4b	All paths	Γ for every path, not just critical. Parasitic paths cause asymmetries.	—
DR-T.5	$M < 10$	Conductance matching.	(8)
DR-T.6	Iterate	Recalculate after improvement.	—
DR-T.7	3σ	Manufacturing robustness.	—
DR-T.8	$\tau_{\text{th}} < \tau_{\text{proc}}$	Temporal compatibility.	—

16 DR Numbering Legacy Map

Table 11: DR numbering legacy map.

Prefix	v3.0 Ch.	Origin	Topic
DR-T.n	4	v2.2 Ch. 4	Topological
DR-05.n	5	v2.2 Ch. 5	Transform/EMT
DR-06.n	5	v2.2 Ch. 6	Materials
DR-09.n	7	v2.2 Ch. 9	Via fabrication
DR-10.n	8	v2.2 Ch. 10	Spectral eng.
DR-09.n	9	v3.0	Multiphysics
DR-11.n	11	v3.0	Semiconductor
DR-13.n	13	v3.0	Manufacturing
DR-14Q.n	14	v3.0	Quantum

DR-09.n appears in both Ch. 7 (via fab) and Ch. 9 (multiphysics). Context disambiguates.

17 Cross-Reference Guide (v4.1)

Table 12: MPM applications across all 15 chapters.

Ch.	MPM Application
1	Fourier–Snell bridge; thermal $\rightarrow$ GRIN $\rightarrow$ WFE.
2	Dynamic $\Gamma(t)$; DR-T.8 temporal compatibility.
3	Near-field G_{rad} enhancement; DR-T.1c.
4	Home chapter. Full derivation, WE-4.0, six-step workflow, DR-T rules.
5	Transformation devices modify G ; EMT replaces microstructure.
6	Doublets reconfigure topology; diodes add nonlinear G .
7	Via arrays as high- G edges; clustered vias to hotspots.
8	G_{rad} as network element; spectral selectivity modifies ε .
9	Mixed-mode ($0.1 < \Gamma < 10$); Alg. 09.A; edge schema DR-09.8.
10	Radiation-dominated conversion; BSR self-consistent iteration.
11	Plasma $\rightarrow$ wafer $\rightarrow$ chuck $\rightarrow$ coolant; dynamic $\Gamma(t)$; thermal Zernike.
12	Critical path shift (CS-5); mode mis-ID (CS-6); I/M/F/T framework.
13	S_i coefficients; tolerance; metrology; observer effect DR-13.10.
14	Dual MPM; $\Gamma \rightarrow 0$ at mK except viewports. DR-14Q.1.
15	Adaptive networks; ML-optimized topology; differentiable MPM.

Annotated References

Table 13: Key references.

Reference	MPM Connection
Kirchhoff (1847)	TKCL/TKVL foundation.
Incropera et al.	R -network; “phonon-only” case.
Modest, <i>Rad. Heat Transfer</i>	Radiosity; “photon-only” case.
Howell, Menguc, Siegel	Enclosure analysis; MPM radiation columns.
Fan & Huang (2008)	Transformation thermotics; Part II.
Xu & Huang (2023)	Diffusionics framework; Part II.
Kelley & Walker (1959)	Critical path method.
Cauer (1926)	RC synthesis; $\mathbf{G} \cdot \mathbf{T} = \mathbf{Q}$.

Book 3 of the Open-Source Engineering Optics Trilogy

Cross-references: Eikonal Bridge (ray-wave duality in propagation);

Quantum Field Imaging (field correlations in detection)